

Graph-Attention Surrogates for Variational Circuit Selection Across Molecular Geometries and Correlation Regimes

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Abstract

Selecting expressive variational circuits without exhaustively optimizing every candidate is a central bottleneck in VQE-based quantum chemistry and correlated-electron simulation. We ask whether graph-attention surrogates can rank particle-number-preserving ansätze when both the circuit architecture and Hamiltonian parameter are unseen.

We benchmark four-qubit half-filled two-site Fermi–Hubbard Hamiltonians and six-qubit active-space LiH in STO–3G. Full runs comprise 1,080 Hubbard and 810 LiH VQE labels with paired optimizer initializations. A lossless Pauli-factor graph retains every aggregated Pauli word, axis, and coefficient; circuit graphs encode excitation order and wires. Edge-conditioned GAT models with MLP or KAN heads learn within-Hamiltonian standardized energies under fingerprint-disjoint splits:

$$a^*(H) = \arg \min_a \min_{\theta} \langle \psi_a(\theta) | H | \psi_a(\theta) \rangle, \quad \Delta E_a = E_a^{\text{VQE}} - E_0.$$

For Hubbard, one simultaneous-transfer split reached Kendall $\tau_b = 0.569$, but the five-seed mean was only 0.180 (SD 0.415, bootstrap 95% CI $[-0.120, 0.540]$), revealing substantial seed sensitivity. On unseen LiH circuits and geometries, circuit GAT led ($\tau_b = 0.259, \rho = 0.366$); factor-GAT+KAN was close ($\tau_b = 0.233, \rho = 0.362$), while Hamiltonian encodings provided no clear advantage. The median LiH gap exceeded 1.6 mHa from 2.4 Å, whereas the best sampled circuit stayed below this threshold throughout. High-precision finalist reruns changed LiH gaps by at most 3.13×10^{-6} Ha. Together, these results show that graph-attention surrogates can support variational-circuit selection across molecular and correlated-electron Hamiltonians, while seed sensitivity makes leakage-controlled validation essential. By screening ansätze before costly VQE optimization, the framework could reduce the number of full optimizations required across molecular geometry scans and correlated-electron parameter sweeps—repeated tasks in quantum-chemical potential-energy surfaces and materials-relevant electronic-structure workflows.

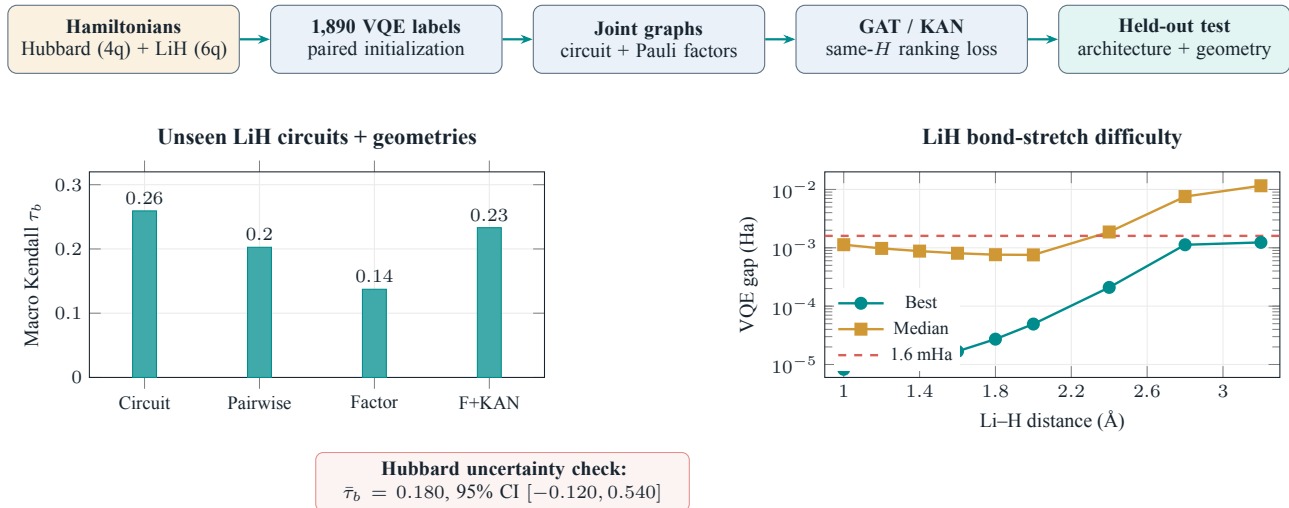


Figure 1. Leakage-controlled QAS workflow and full-mode LiH results. The dashed line is a numerical active-space threshold, not an experimental-error claim.

Keywords: quantum architecture search; variational quantum eigensolver; graph attention network; electronic structure; quantum chemistry

Data and code. S. M. Hoque, *Graph-Attention-QAS*, 2026, github.com/SyedT1/Graph-Attention-QAS.